

Cyber Risk and Framework Compliance

Indian Institute of Technology / Apex University - scan of 2026-08-28 - 18 assets - 18 findings

EXPECTED ANNUAL LOSS	RISK INDEX (0-100)	CRITICAL / KEV FINDINGS	REQUIREMENTS CITED
INR 68,57,300	62	13 / 12	90

Basis of the figures in this report

Model state	HARNESS FIXTURE: no trained artefact is loaded in this check.
Data	The loaded artefact is NOT built from real data. Treat every figure below as a pipeline demonstration, not as a measurement of this organisation.
Coverage	9 of 18 findings were scored by the model. The remainder use the deterministic fallback, which yields a score rather than a calibrated probability.
Artefact built	No model artefact is loaded; all figures are from the deterministic fallback.
Parity check	Not run - the artefact shipped no parity fixture, so browser-side scoring has not been confirmed against the Python training code.
Framework catalogue	Built 2026-09-01T17:51:59+00:00. Identifier verification is defined on the closing page.
- Figures below are deterministic placeholders produced by scripts/check-exports.cjs.	

Coverage of the five frameworks

Cited controls are those this scan produced evidence about. The rest are not assessed; this report makes no claim that they are implemented.

Framework	Authority	Cited	In scope	Planned	Open	Weakest ID status
ISO/IEC 27001:2022	International Organization for Standardization	22 / 54	40.7%	17	5	verified
NIST CSF 2.0	National Institute of Standards and Technology	18 / 64	28.1%	15	3	verified
CIS Controls v8.1	Center for Internet Security	29 / 81	35.8%	17	12	verified
RBI Cyber Security Framework	Reserve Bank of India	11 / 30	36.7%	8	3	inferred
SEBI CSCRF	Securities and Exchange Board of India	10 / 25	40%	8	2	assigned

Where the exposure sits

The requirements with the most expected annual loss behind them. Attributed exposure is not additive down this column: one finding is evidence against several controls, so the same rupee appears on every row it implicates. The portfolio total is on the first page.

Framework	Control	Requirement	Findings	Attributed EAL	Remediation
ISO/IEC 27001:2022	A.8.8	Management of technical vulnerabilities	18	INR 68,57,300	CTRL-05, CTRL-08
NIST CSF 2.0	ID.RA-01	Vulnerabilities in assets are identified, validated and recorded	18	INR 68,57,300	CTRL-08
NIST CSF 2.0	ID.RA-05	Threats, vulnerabilities, likelihoods and impacts are used to understand inherent risk and inform risk response prioritization	18	INR 68,57,300	CTRL-08
CIS Controls v8.1	7.1	Establish and maintain a vulnerability management process	18	INR 68,57,300	CTRL-08
CIS Controls v8.1	7.5	Perform automated vulnerability scans of internal enterprise assets	18	INR 68,57,300	CTRL-08

Framework	Control	Requirement	Findings	Attributed EAL	Remediation
CIS Controls v8.1	7.7	Remediate detected vulnerabilities	18	INR 68,57,300	CTRL-05, CTRL-08
RBI Cyber Security Framework	RBI-A1-07 (inferred)	Patch / vulnerability and change management	18	INR 68,57,300	CTRL-05, CTRL-08
SEBI CSCRF	SEBI-ID-05 (assigned)	Vulnerability assessment and penetration testing (VAPT)	18	INR 68,57,300	CTRL-08
SEBI CSCRF	SEBI-PR-04 (assigned)	Patch and vulnerability remediation management	18	INR 68,57,300	CTRL-05, CTRL-08
SEBI CSCRF	SEBI-DE-03 (assigned)	Continuous threat detection and alerting	9	INR 48,94,200	CTRL-04, CTRL-06
ISO/IEC 27001:2022	A.8.26	Application security requirements	9	INR 46,46,300	CTRL-05
ISO/IEC 27001:2022	A.8.28	Secure coding	9	INR 46,46,300	none mapped
NIST CSF 2.0	PR.PS-06	Secure software development practices are integrated, and their performance is monitored throughout the software development life cycle	9	INR 46,46,300	none mapped
CIS Controls v8.1	16.12	Implement code-level security checks	9	INR 46,46,300	none mapped

Largest single exposures

CVE	Asset	Criticality	CVSS	KEV	Figures from	Expected annual loss
CVE-2023-38606	Student Information & Records DB	Critical	9.8	yes	model	INR 11,12,500
CVE-2024-3400	Campus Core Gateway & Perimeter Firewall	Critical	10.0	yes	model	INR 10,34,000
CVE-2024-21413	Central Active Directory & SSO IdP	Critical	9.8	yes	heuristic	INR 7,98,000
CVE-2023-46604	Online Examination & Evaluation Portal	Critical	9.8	yes	model	INR 7,28,000
CVE-2023-34362	University Payment & Fee Gateway	Critical	9.8	yes	heuristic	INR 5,32,000
CVE-2023-3519	Virtual Desktop Infrastructure (VDI Farm)	High	9.8	yes	model	INR 3,61,200
CVE-2024-1086	Faculty Research Supercomputing Cluster (HPC)	High	7.8	yes	model	INR 3,48,000
CVE-2022-39952	Enterprise ERP & Faculty Payroll System	High	9.8	yes	heuristic	INR 3,40,000
CVE-2023-7028	Internal Code Repository (GitLab CE)	High	10.0	yes	heuristic	INR 3,00,200
CVE-2023-50164	Learning Management System (LMS - Moodle)	High	9.8	no	heuristic	INR 2,92,500

Risk metrics against reporting requirements

4 of 8 metrics are computable from this scan. A metric shown as unavailable has not been measured; it is not a zero, and must not be reported as one.

Metric	This scan	Requirement	Governance statement
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Metric	This scan	Requirement	Governance statement
Expected annual loss	INR68,57,300 per year	12	Quantified financial consequence of the current open finding set, used to demonstrate that risk is assessed in business terms and against a stated appetite rather than by severity label alone.
Exploitation probability (model output)	not available - harness: no model is loaded in this check	11	Threat-informed likelihood per finding. Evidences that prioritisation uses current threat intelligence and not just static CVSS severity.
Residual risk after planned mitigation	not available - No mitigation portfolio has been selected yet, so residual risk is undefined.	13	The number a board must formally accept. Reporting it alongside the pre-treatment figure is what turns a scan into a risk-acceptance record.
Framework control coverage	72.2% of 90 implicated controls have a mapped remediation	13	Proportion of mapped controls with at least one satisfying investment or no contradicting finding. This is the self-assessment percentage regulators ask for, and for SEBI-regulated entities it is the input to the Cyber Capability Index submission.
Critical and high finding count	13	11	Open severe exposures at reporting date, with ageing. Directly evidences whether remediation timelines are being met.
Findings under active exploitation	12	12	Count of findings matching a confirmed in-the-wild exploitation catalogue. Usually the single most escalation-worthy line in a board pack.
Return on security investment	not available - ROI requires a selected investment portfolio; none has been optimised yet.	8	Risk reduction per rupee for the selected portfolio. Evidences that resourcing decisions are risk-based, which is a governance obligation in its own right and not merely a finance question.
Business-criticality weighted exposure	not available - No assets carry a business-criticality label in this scan.	13	Shows the asset inventory is classified and that risk is weighted by business impact - the precondition for every other number here being meaningful.

Recommended portfolio at the stated budget

Budget	INR 50,00,000, of which INR 55,00,000 is allocated (74%).
Risk index	62 today, 48 after this portfolio.
Exposure	INR 68,57,300 per year today, INR 42,51,526 after. Avoided: INR 26,05,774 per year.
Return on security spend	210% average, computed as avoided annual loss over cost.
Selection rule	Greedy by risk reduction per rupee, then compounded with diminishing returns. This is a ranking heuristic, not a proof of optimality.

Control	Investment	Cost	Reduction	Requirements it would satisfy
CTRL-01	Deploy Zero-Trust Microsegmentation	INR 18,00,000	32%	A.8.20, A.8.21, DE.CM-01, PR.AA-05, 4.4, 12.2, RBI-A1-04, RBI-A1-05, SEBI-DE-01, SEBI-PR-03
CTRL-02	Phishing-Resistant FIDO2/Hardware MFA	INR 12,00,000	28%	A.5.15, A.5.16, PR.AA-01, PR.AA-02, 5.2, 6.3, RBI-A1-08, RBI-A1-09, SEBI-PR-01, SEBI-PR-02
CTRL-03	Database Activity Monitoring (DAM) & Encryption	INR 15,00,000	25%	A.5.12, A.8.3, DE.CM-03, PR.DS-01, 3.3, 3.10, RBI-A1-14, RBI-A1-15, SEBI-DE-02, SEBI-PR-05
CTRL-04	AI-Driven Managed XDR Endpoint Detection	INR 10,00,000	22%	A.5.7, A.5.24, DE.AE-02, DE.AE-03, 8.2, 8.5, RBI-A1-19, RBI-A1-20, SEBI-DE-01, SEBI-DE-02

How to read this report

Identifier verification

verified (69 cited)	Identifier and title checked against the published standard. Citable as a reference to that standard.
inferred (11 cited)	Requirement substance follows the primary document, but the official clause numbering could not be confirmed. NOT a citation.
assigned (10 cited)	Identifier created by this platform, with the mapping basis recorded on each row. NOT a citation.

verification=verified means the identifier and title were checked against the published standard. inferred means the requirement substance is correct but the official clause numbering could not be confirmed from the primary document. assigned means this project created the identifier and records the basis for the mapping. Reports must surface this status; an inferred or assigned identifier is not a citation.

Attributed exposure is not additive

Attributed EAL is not additive. One finding implicates several controls, so the same rupee of expected annual loss appears on every row it is evidence for. Use the portfolio total on the Summary sheet, never a column sum.

The portfolio total for this scan is INR 68,57,300 per year. Any larger figure obtained by summing a control column is double-counting.

What silence means

A control this report does not cite is one the scan produced no evidence about. That is not a statement that the control is implemented, and it cannot be used as one. Framework coverage figures describe how much of each framework this scan can speak to, not a compliance score.

Known gaps in this run

Unclassified findings	None. Every finding resolved to a vulnerability class and implicates at least one control.
Unmapped investments	None. Every candidate investment maps to at least one framework requirement.

Sources consulted

ISO/IEC 27001:2022	International Organization for Standardization. ISO/IEC 27001:2022, Annex A. Control identifiers and titles as published by ISO/IEC.
NIST CSF 2.0	National Institute of Standards and Technology. NIST Cybersecurity Framework 2.0 (NIST CSWP 29, February 2024). Subcategory identifiers as published by NIST.
CIS Controls v8.1	Center for Internet Security. CIS Critical Security Controls v8.1 (Center for Internet Security). Safeguard numbering as published by CIS.
RBI Cyber Security Framework	Reserve Bank of India, 'Cyber Security Framework in Banks', circular DBS.CO/CSITE/BC.11/33.01.001/2015-16 dated 2 June 2016 (Annex 1 baseline requirements, Annex 2 C-SOC, Annex 3 CCMP). Internal RBI-A<n>-<nn> identifiers are this project's numbering; requirement substance follows the circular.
SEBI CSCRF	Securities and Exchange Board of India. SEBI, 'Cybersecurity and Cyber Resilience Framework (CSCRF) for SEBI Regulated Entities', circular SEBI/HO/ITD-1/ITD_CSC_EXT/P/CIR/2024/113 dated 20 August 2024, with the June 2025 FAQ and the 28 August 2025 technical clarifications. Internal SEBI-<FN>-<nn> identifiers are this project's numbering, grouped by the CSCRF's NIST CSF 2.0 anchoring.